Automatic recognition of the buḥūr

Classical Arabic meter, formalised for an algorithm · Project overview

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https://github.com/BassamSINAN/classical-arabic-meter-modeling

An old rule system that is still hard to automate

The ʿarūḍ, formalised by al-Ḫalīl ibn Aḥmad al-Farāhīdī, describes classical Arabic poetry with sixteen meters (buḥūr). Once its rules are learnt, the system is regular and predictable.

The question: can a system recognise the meter of a verse automatically, taking its variations into account, while staying transparent, reproducible and open?

Partial coverage

Existing tools (Xaliyl, BASRAH, among others) handle a limited set of meters or skip some variations

Opaque rules

Their algorithms, rules and exceptions are rarely documented

No open code

No open-source system was found that can be reused or reproduced

Every verse splits into two matching halves

A verse attributed to ʿAntara ibn Shaddād, pre-Islamic poet, sung by Fayrouz

لو كان قلبي معي ما اخترتُ غيرَكم

ṣadr («chest»): first hemistich

ولا رضيتُ سِوَاكُمْ في الهَوى بَدَلا

ʿağuz («rear»): second hemistich

← *read right to left*

- A poem (qaṣīda) is a sequence of verses (bayt, pl. abyāt).
- Each bayt has two hemistichs (šaṭr) that share, in principle, the same metrical pattern.
- This symmetry is what the algorithm will check: same meter in both halves.

From text to meter in four steps

1 → 2 → 3 → 4

kitāba ʿarūḍiyya

Prosodic transcription

Write the verse as it is heard, not as it is spelled

taqṭīʿ

Scansion

Cut it into short and long segments (maqāṭiʿ)

tafāʿīl

Metrical feet

Group the segments into feet, allowing the permitted variations

baḥr

Meter

Match the sequence of feet against the sixteen meters

Where the difficulty lies: matching feet to a meter is simple once the scansion is right. Most of the complexity sits in steps 1 and 2.

Step 1 · Write what is heard, not what is spelled

Rule 1: what is pronounced is written

Doubled consonant (šadda)

عَدَّ ← **عَدْدَ**

Nunation (tanwīn) becomes a nūn

جَبَلٌ ← **جَبَلُنْ**

Alif heard but not spelled

هَذَا ← **هَاذَا**

Pronoun vowel lengthened

لَهُ ← **لَهُو**

Rule 2: what is not pronounced is not written

Silent hamzat al-waṣl

طَلَعَ القَمَرُ ← **طَلَعَ لْقَمَرُ**

Article assimilated (šamsī)

طَلَعَتِ الشَّمسُ ← **طَلَعَتِ شْشَمسُ**

Two sākin letters cannot meet

فِي الْبَحْرِ ← **فِلْبَحْرِ**

Silent alif after verb plural

كتبوا ← **كتبو**

The two rules follow Abū Ṭālib (2017). In the code, each case is a separate rule with its own tests.

Step 2 · Cut the verse into short and long segments

u

maqṭaʿ qaṣīr · short

One moving letter (mutaḥarrik): a consonant with a short vowel

عَ / لِ / مَ

عَلِمَ = u u u

–

maqṭaʿ ṭawīl · long

A moving letter followed by a resting one (sākin)

مَا · لَمْ · قَطْ · كُنْ

each = –

Notation in this deck

Segment	Arabic tradition	Paoli (2008), used here	Abū Ṭālib (2017)
short	ن	u	/
long	-	–	o/

Step 3 · Eight feet built from small blocks

sabab 2 letters

ḫafīf: – · ṯaqīl: u u

watad 3 letters

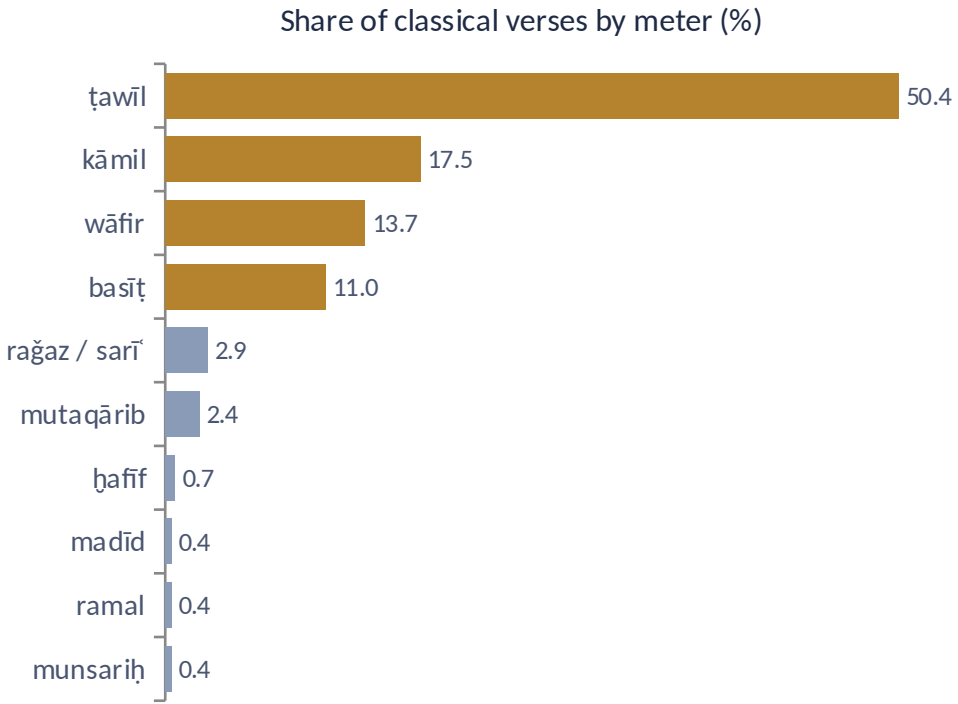
maǧmūʿ: u – · mafrūq: – u

fāṣila 4–5 letters

ṣuġrā: u u – · kubrā: u u u –

Foot		Pattern
فَاعِلُن	*fāʿilun*	– u –
فَاعِلَاتُن	*fāʿilātun*	– u – –
مُسْتَفْعِلُن	*mustafʿilun*	– – u –
مَفْعُولَاتُ	*mafʿūlātu*	– – – u
فَعُولُن	*faʿūlun*	u – –
مَفَاعِيلُن	*mafāʿīlun*	u – – –
مُفَاعَلَتُن	*mufāʿalatun*	u – u u –
مُتَفَاعِلُن	*mutafāʿilun*	u u – u –

Step 4 · Sixteen meters, but four cover 93% of verses



93%

of verses use ṭawīl, kāmil, wāfir or basīṭ

Scope of the thesis: these four meters. A meter is recognised by matching its feet against its key pattern (miftāḥ).

Source: Golston & Riad (1997), pp. 120–121. Five other meters each under 0.43%.

The real difficulty: feet change shape

Position of each foot in a verse of three feet per hemistich (reading order)

ṣadr			*ʿağuz*		
ḥašw	*ḥašw*	*ʿarūḍ*	*ḥašw*	*ḥašw*	*ḍarb*

zihāfāt on inner feet (ḥašw)

Optional. They touch the sabab, and a poet may use them in one verse and not the next.

ʿilal on final feet (ʿarūḍ, ḍarb)

Fixed by the first verse and kept for the whole poem.

Example: mustafʿilun – – u – loses its second letter (the sākin s) and becomes mutafʿilun u – u –.
So comparing a verse with the canonical pattern is not enough: the algorithm must list every valid form of each foot, by position and by meter.

ʿAntara's verse, step by step: it is basīṭ

ṣadr

لو كان قلبي معي ما اخترتُ غيرَكم

law kā na qal	*bī ma ʿī*	*makh tar tu ġay*	*ri ku mū*
– – u –	– u –	– – u –	u u –
mustafʿilun	fāʿilun	mustafʿilun	faʿilun *

ʿağuz

ولا رضيتُ سِوَاكُمْ في الهَوى بَدَلا

wa lā ra ḍī	*tu si wā*	*kum fil ha wā*	*ba da lā*
u – u –	u u –	– – u –	u u –
mutafʿilun *	faʿilun *	mustafʿilun	faʿilun *

Key of basīṭ: mustafʿilun fāʿilun mustafʿilun fāʿilun. * = permitted variation (the second letter, a sākin, drops). Transcription: mā iḫtartu → maḫtartu, ġayrakum → ġayrakumū, fī l-hawā → fil-hawā.

Where the project stands

1

Data *Done*

Ashaar corpus cleaned: 265,609 verses, of which 72,654 in ṭawīl. Verses kept only if well vocalised.

2

Rule-based algorithm f *In progress*

Prototype: recognises ʿAntara's verse and 112 of 160 test verses. Transcription: 15 rules, 63 tests.

3

Learning model *Planned*

A machine-learning model trained on the errors of f, to handle its exceptions.

4

Web interface *Later*

Scan a verse online.

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